ROTARY MACHINERY MAINTENANCE STANDARD
Document No: P311-DOC-PUMP-COMP-01
Scope: Centrifugal Pumps (MOTOR-PUMP-01) & Twin-Screw Compressors (MOTOR-COMP-03)

SECTION 1: ROTATING EQUIPMENT OPERATING ENVELOPE
Centrifugal Coolant Pump (MOTOR-PUMP-01):
- Rated Power: 15 kW, 28A FLA, Nominal Speed: 1450 RPM.
- Normal discharge pressure: 4.0 to 4.5 bar.
- Thermal casing limit: Warning at 75°C, Critical emergency trip at 90°C.
- Mechanical vibration: ISO 10816 Zone C warning at 4.5 mm/s, Zone D trip at 7.1 mm/s.

Air Compressor (MOTOR-COMP-03):
- Rated Power: 22 kW, 41A FLA, Nominal Speed: 2960 RPM.
- Operating delivery pressure: 7.0 to 8.0 bar.
- Oil separator temperature: Normal 65°C to 72°C. Critical trip at 92°C.

SECTION 2: HYDRAULIC & MECHANICAL FAILURE MODES

2.1 Pump Cavitation & Impeller Degradation:

Cavitation is identified by audible crackling (above 85 dB) accompanied by an unstable discharge pressure dropping below 3.0 bar. Impeller pitting will occur within 48 hours of continuous cavitation.

Corrective Steps:

1. Verify suction line strainer is free of particulates.
2. Ensure pump priming chamber is completely vented of entrained air.
3. Check NPSH available exceeds manufacturer requirement by at least 0.5 meters.

2.2 Compressor High-Speed Bearing Friction:

High rotational speed (2960 RPM) generates rapid thermal gradients when lubrication degrades. Vibration frequency peaks at 1X and 2X indicate coupling misalignment or rotor balance drift. Torque hold-down foundation bolts to 85 Nm.

SECTION 3: ELECTRICAL OVERLOAD & POWER QUALITY
3.1 Overcurrent Diagnostics:
- Normal Amperage: 85-95% of full-load rating.
- Warning Threshold: 115% FLA (32.2A on pump, 47.1A on compressor).
- Critical Trip Limit: 135% FLA (37.8A on pump, 55.3A on compressor).
Causes of overcurrent include driven load mechanical jamming, slurry thickening, or supply voltage sag below 207V.

3.2 OSHA Lockout-Tagout (LOTO) Compliance:
Always de-energize the motor control center (MCC) breaker, insert personal lockout hasp, and verify zero mechanical and electrical potential prior to opening junction box or coupling guard.